SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 10%

11. An image contains high-frequency noise. Explain and justify the appropriate frequency domain filter, including its working and impact on image quality.
12. You need to preserve low-frequency components while removing noise. Describe and apply a suitable filtering technique, explaining how it achieves the desired effect.
13. An image requires edge enhancement in the frequency domain. Explain and justify the filtering approach used, describing its effect on frequency components.
14. A smooth version of an image is required. Describe and justify the frequency domain technique used for smoothing, explaining its operation and results.
15. An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.
16. You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.
17. An image requires identification of object edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.
18. Adjacent pixels have similar intensity values. Describe and apply a segmentation technique to group such pixels, explaining the concept and reasoning.
19. Detected edges in an image appear broken. Explain and justify a method to improve edge continuity, including its role in post-processing.

20. You need to extract object outlines from an image. Describe and apply an appropriate technique, explaining how it identifies boundaries effectively.

192425005

CLASS TEST

ITA0504 – Computer Vision

1) Image Appears Too Bright

Technique: Negative Transformation Negative transformation converts bright pixels into dark pixels using.

$$\boxed{s = L - 1 - r}$$

85%

* It reduces excessive brightness, improves visibility, and reveals hidden details in overexposed regions.

2) Dark Image Lacks Details

Technique: Log Transformation Expands low-intensity values and compress high-intensity values using

$$\boxed{s = c\log(1 + r)}$$

* It brights dark areas and makes hidden details more visible.

3) Poor Contrast Image

Technique: Constrast Stretching increases the intensity range by mapping pixel values to full gray scale (0–255). This improves contrast and makes object easier to distinguish.

4) Uniform Intensity Distribution Needed.

Technique: Histogram Equalization redistributions pixel intensities uniformly using the cummulative histogram. It enhances global contrast and improves overall image visibility.

5) Uneven Brightness Across Regions

Techniques: CLAHE (Adaptive Histogram Equalisation) enhances small regions separately while limiting over-enhancement. It corrects uneven illumination and improves local contrast without amplifying noise.

6) Random Intensity Noise:

Techniques: Median Filter replaces each pixel with the median of its neighbouring pixels. It effectively removes salt-and-pepper noise while preserving image edges.

7) Noise Reduction Without Losing Details

Technique: Bilateral Filter smooths the image using both spatial distance and intensity similarity. It reduces noise while preserving edges and fine details.

Working Process:

i) Compute weights based on pixel distance.

ii) Compute weights based one intensity similarity.

iii) Combine both weights.

iv) Replace each pixel with the weighted average.

Justification:

* Removes noise.
* Preserves edges and fine details.
* Produces natural-looking results.
* Sharp edges retained.
* High-quality enhanced images.

8) Blurred Image After Smoothing

Suitable Sharpening Method: Unsharp Masking.

Concept:

Unsharp masking sharpens an image by adding back the high-frequency details removed during smoothing.

Working Principle:

1. Blur the original Image.
2. Subtract the blurred image from the original.
3. Add the difference back to the original image.

Justification:

* Restores lost details.
* Enhances edges.
* Improves Image sharpness.

Expected Outcome:

* Sharper Image.
* Better edge definition.
* Improved visual clarity.

9) Fine details Not Clearly Visible

Suitable filtering Technique: High-Pass filter.

Concept:

A high-pass filter enhances high-frequency components such as edges and fine details.

Application Process:

1. Apply a high-pass kernal.
2. Suppress low-frequency components.
3. Highlight rapid intensity enhances.
4. Produce the enhanced image.

Justification:

* Emphasizes edges.
* Reveals fine textures.
* Improves feature visibility.

Expected Outcome:

* Enhanced details.
* Sharper edges.
* Better features extractions.

10) Noisy Image Before Analysis

Technique : Gaussian Filter

The Gaussian filter smooths the image using weighted averaging to reduce Gaussian noise. It prepares the image for further processing while maintaining a natural appearances.

Code of Conduct:

I S. Yuva Guru Mani Varma /192425005 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *S. Yuva Guru Mani Varma*

Rubrics for Evaluation

Criteria	*Weightage*	*Level 4 – Exemplary*	*Level 3 – Proficient*	*Level 2 – Developing*	*Level 1 – Beginning*
Understanding of Concepts	*25%*	*Demonstrates thorough, accurate understanding of concepts*	*Good understanding with minor gaps*	*Basic understanding with some misconceptions*	*Poor understanding; major misconceptions*
Explanation	*25%*	*Detailed, well-explained answers with relevant examples*	*Clear explanation with adequate details*	*Limited explanation; lacks depth*	*Poor or unclear explanation*
Application	*20%*	*Effectively applies concepts with strong analytical ability*	*Applies concepts with minor errors*	*Limited application with weak analysis*	*No application or incorrect analysis*
Organization	*15%*	*Well-structured answers with logical flow and coherence*	*Organized with minor issues in flow*	*Basic structure, lacks clarity*	*Poor organization, incoherent answers*

Accuracy	*10%*	*Completely accurate and covers all required points*	*Mostly accurate with minor omissions*	*Partially correct with missing points*	*Incorrect answers with major gaps*
Clarity	*5%*	*Clear, neat, professional presentation*	*Understandable with minor issues*	*Limited clarity, readability issues*	*Poorly written, difficult to understand*